

Drive for better vision



UART OTA User Guide V1.2

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Agenda

- **Check Before Starting OTA**
- **OTA Upgrade Operations**

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Check Before Starting OTA

Check Before Starting OTA

- Generate FW image with UART protocol
 - ❖ Set TRANSMIT_PROTOCOL to UART in sample_code_app app_main.mk

```
#####  
## Transmit Protocol Option ##  
## - UART                      ##  
## - SPI_MASTER                ##  
## - SPI_SLAVE                 ##  
#####  
TRANSMIT_PROTOCOL ?= UART
```

- ❖ In SDK Root, use menuconfig to configure 2nd bootloader

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184116_ISM028_UART_OTA>make menuconfig
```

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- ❖ In menuconfig, select “1. Loader Config”

```
[HIMAX] *****  
[HIMAX] *****  
[HIMAX] ***** MenuConfig Ver.1.0.0 *****  
[HIMAX] *****  
[HIMAX] *****  
[HIMAX] Feature list:  
[HIMAX] 1. Loader Config (Option)  
[HIMAX] 2. Return Without Saving  
[HIMAX] 3. Save and Exit  
[HIMAX] 4. *** End of List ***  
[HIMAX]  
[HIMAX] Please select one feature ( 1 ~ 3 ): 1
```

- ❖ Select “2. Second Boot Loader”

```
[HIMAX] ----- Boot Loader List -----  
[HIMAX] 1. First Boot Loader  
[HIMAX] 2. Second Boot Loader  
[HIMAX] 3. <BACK>  
[HIMAX] 4. *** End of List ***  
[HIMAX]  
[HIMAX] Please select loader config ( 1 ~ 3 ): 2
```

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- ❖ Select “4. Command UART(1) Pin Mux”

```
[HIMAX] ----- Second Boot Loader -----  
[HIMAX] 1. Clock Case  
[HIMAX] 2. Clock Type  
[HIMAX] 3. Other Flag  
[HIMAX] 4. Command UART(1) Pin Mux  
[HIMAX] 5. Console Baud Rate  
[HIMAX] 6. Command Baud Rate  
[HIMAX] 7. CPU Frequency  
[HIMAX] 8. <BACK>  
[HIMAX] 9. *** End of List ***  
[HIMAX]  
[HIMAX] Please select one option ( 1 ~ 8 ) 4
```

- ❖ Select “2. PB6(RX), PB5(TX)” for ISM028 with DB2.

```
[HIMAX] ----- 2nd Boot Loader Cmd UART1 Pin Mux List -----  
[HIMAX] 1. PB4(RX), PB5(TX)  
[HIMAX] 2. PB6(RX), PB5(TX)  
[HIMAX] 3. PB6(RX), PB7(TX)  
[HIMAX] 4. PB4(RX), PB7(TX)  
[HIMAX] 5. <BACK>  
[HIMAX] 6. *** End of List ***  
[HIMAX]  
[HIMAX] Please select options ( 1 ~ 5 ) 2
```

Check Before Starting OTA

- ❖ Select “6. Command Baud Rate”

```
[HIMAX] ----- Second Boot Loader -----  
[HIMAX] 1. Clock Case  
[HIMAX] 2. Clock Type  
[HIMAX] 3. Other Flag  
[HIMAX] 4. Command UART(1) Pin Mux  
[HIMAX] 5. Console Baud Rate  
[HIMAX] 6. Command Baud Rate  
[HIMAX] 7. CPU Frequency  
[HIMAX] 8. <BACK>  
[HIMAX] 9. *** End of List ***  
[HIMAX]  
[HIMAX] Please select one option ( 1 ~ 8 ): 6
```

- ❖ Enter the Command Baud Rate “921600”

```
[HIMAX] ----- Second Boot Loader -----  
[HIMAX] 1. Clock Case  
[HIMAX] 2. Clock Type  
[HIMAX] 3. Other Flag  
[HIMAX] 4. Command UART(1) Pin Mux  
[HIMAX] 5. Console Baud Rate  
[HIMAX] 6. Command Baud Rate  
[HIMAX] 7. CPU Frequency  
[HIMAX] 8. <BACK>  
[HIMAX] 9. *** End of List ***  
[HIMAX]  
[HIMAX] Please select one option ( 1 ~ 8 ): 6  
[HIMAX] Your selection is "Command Baud Rate"  
[HIMAX] ----- OTA CMD Baud Rate List -----  
[HIMAX] Please enter OTA cmd baud rate (or press ENTER to use default: 921600) 921600
```

Check Before Starting OTA

- ❖ Select “8. <BACK>”

```
[HIMAX] ----- Second Boot Loader -----  
[HIMAX] 1. Clock Case  
[HIMAX] 2. Clock Type  
[HIMAX] 3. Other Flag  
[HIMAX] 4. Command UART(1) Pin Mux  
[HIMAX] 5. Console Baud Rate  
[HIMAX] 6. Command Baud Rate  
[HIMAX] 7. CPU Frequency  
[HIMAX] 8. <BACK>  
[HIMAX] 9. *** End of List ***  
[HIMAX]  
[HIMAX] Please select one option ( 1 ~ 8 ): 8
```

- ❖ Select “3. <BACK>”

```
[HIMAX] ----- Boot Loader List -----  
[HIMAX] 1. First Boot Loader  
[HIMAX] 2. Second Boot Loader  
[HIMAX] 3. <BACK>  
[HIMAX] 4. *** End of List ***  
[HIMAX]  
[HIMAX] Please select loader config ( 1 ~ 3 ): 3
```


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- ❖ Select “3. Save and Exit” and the settings will be updated.

```
[HIMAX] *****  
[HIMAX] *****  
[HIMAX] ***** MenuConfig Ver.1.0.0 *****  
[HIMAX] *****  
[HIMAX] *****  
[HIMAX] Feature list:  
[HIMAX] 1. Loader Config (Option)  
[HIMAX] 2. Return Without Saving  
[HIMAX] 3. Save and Exit  
[HIMAX] 4. *** End of List ***  
[HIMAX]  
[HIMAX] Please select one feature ( 1 ~ 3 ): 3
```

- ❖ Build FW image to include the 2nd bootloader settings with UART.

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184116_ISM028_UART_OTA>make clean all flash
```

- Make sure the generated output.img which includes the UART settings is burn to the EVB before doing the OTA operation.

Check Before Starting OTA

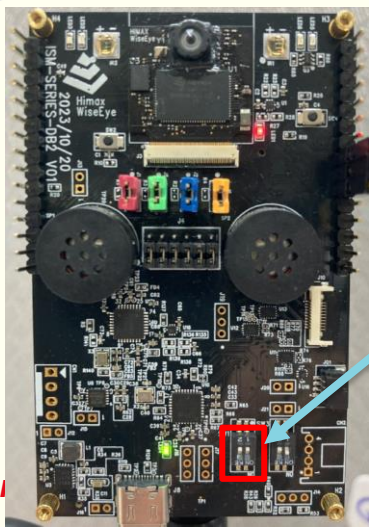
1. Does the EVB already been burned with any firmware?

- ❖ If do, the EVB can upgrade firmware through OTA.
- ❖ If not, the EVB **CANNOT** upgrade firmware through OTA.
To use OTA, there must be any firmware on the EVB so that it can boot up correctly.

To learn more about burning firmware with PC tool,
please refer to “[2_EVK_and_PC_Tool_User_Guide_HX6538_ISM028_03M_V1.2](#)”.

2. Do the EVB and debug board already set to the normal mode?

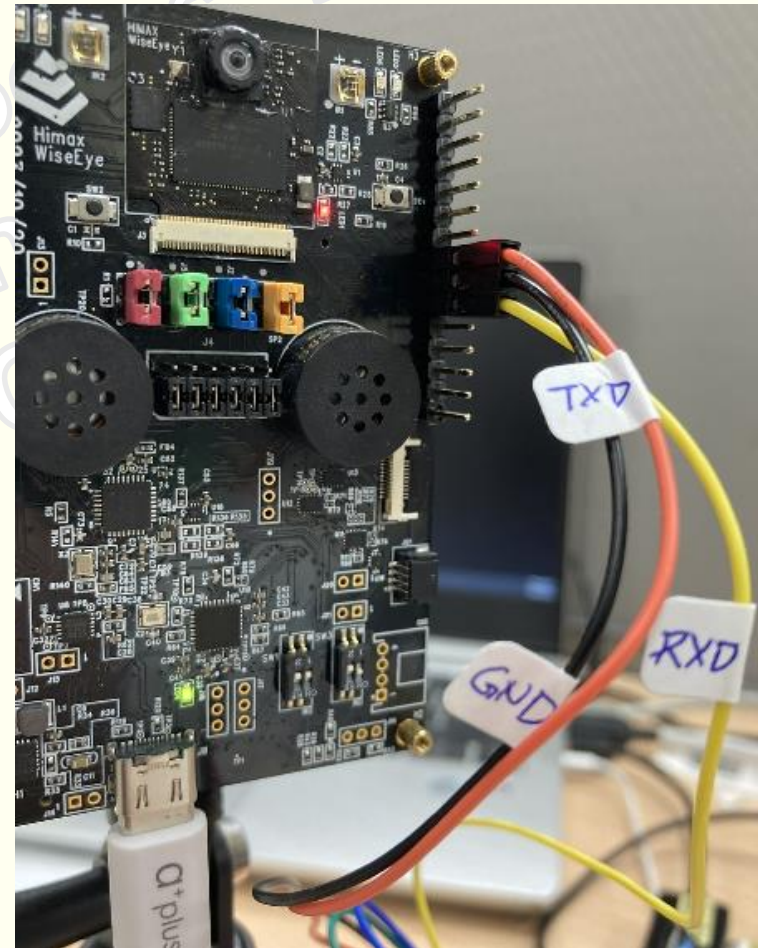
- ❖ If not, two steps to set normal mode:
 - ① Turn off the EVB power.
 - ② Set the switches on EVB: **SWITCH [2, 1] = [OFF, OFF]**



SWITCH SETTING	MODE
 2 1	EVB: Normal mode

Check Before Starting OTA

- Connect additional USB to UART dongle to DB2
 - ❖ UART1_TX: PB5 (J11 #6)
 - ❖ GND (J11 #7)
 - ❖ UART1_RX: PB6 (J11 #8)



OTA Upgrade Operations

OTA Upgrade Operations (1/6)

- Turn on the power of EVB
- Launch OTA tool
(path: SDK_root/we2_ota_tool/ota.exe)
- Operate by following steps:
 - ① **Config the UART for protocol**
 - a) Key in "2" (1. UART protocol)
 - b) Enter the com port number
 - c) Enter the UART baudrate

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Please choice transfer protocol:
1.I2C protocol
2.UART protocol
0.exit

2
3.COM81
Input port number: (ex : 64)
81
1. 300
2. 600
3. 1200
4. 2400
5. 4800
6. 9600
7. 14400
8. 19200
9. 38400
10. 57600
11. 115200
12. 230400
13. 460800
14. 921600
Input baud rate : (ex : 921600)
921600.
```

- ② **Key in "1" (1. Command)**

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : UART
=====OTA Main Menu=====
Please choice following one item:
1.Command
2.Upgrade
3.Others
0.exit

1.
```

OTA Upgrade Operations (2/6)

③ Key in “8” (8. Intoupg)

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb 1 2024=====
Transfer Protocol : UART
=====OTA Command Menu=====
Please choice following one item:
1.Get OTA Loader Version.....[OTA LOADER]
2.Read WE2 Chip id.....[OTA LOADER]
3.Getstatus (Get WE-II current status).....[OTA LOADER]
4.ReBoot WEII.....[OTA LOADER]
5.Busy pin select.....[OTA LOADER]
6.Read busy pin select.....[OTA LOADER]
7.Reserved.....[NONE]
8.Intoupg.....[APPLICATION]
0.exit/back

8

Intoupg
UART_PROTOCOL
uart init
COM81 OPENED BAUDRATE:921600
Press Enter key to continue...
```

Hint: If you also start a log monitor, the console of log will show that:

```
COM72 - Tera Term VT
File Edit Setup Control Window Help
[app_dp_capture_frame]
[app_dp_get_frame] raw_img_addr = 34089f08 img_width = 640 img_height = 480, jpg_
addr = 34082c00, jpg_size = 15076
app_algo_run...
bbox_quantity = 1
app_uart_write(size = 15076, data_type = 1)
app_uart_write(header) return 7
app_uart_write(payload) return 15076
app_uart_write(size = 2041, data_type = 19)
app_uart_write(header) return 7
app_uart_write(payload) return 2041
[MSG] feature_type: 0x50
[MSG] cmd_id: 0x8
[MSG] payload_len: 0x0
===UARTCOMM_CMD OTA_JUMP2UPG===
1st BL Build DATE=Aug 1 2023, 0x00010001
Jump_addr=0x3401f000
Compiler Version: ARM CLANG, Clang 12.0.0 (ssh://ds-gerrit/armcompiler/11vm-proje
ct_e64d644232aba72041b013571dc92e0d3fb7b4e2)
set_IP_secure done
flash type[1], flash size[4]
2ndBootLoader Version : SVN_183742, Feb 6 2024, 14:17:28
Chip Version : 0x8538000c
OTA L/F : I2C UART
SecureLib Version : SVN_181742
Serial number : 0x0
Part number : 0x0
```

OTA Upgrade Operations (3/6)

④ Press any key

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : UART
=====OTA Command Menu=====
Please choice following one item:
1.Get OTA Loader Version.....[OTA LOADER]
2.Read WE2 Chip id.....[OTA LOADER]
3.Getstatus (Get WE-II current status).....[OTA LOADER]
4.ReBoot WEII.....[OTA LOADER]
5.Busy pin select.....[OTA LOADER]
6.Read busy pin select.....[OTA LOADER]
7.Reserved.....[NONE]
8.Intoupg.....[APPLICATION]
0.exit/back

                               8

Intoupg
UART_PROTOCOL
uart init
COM81 OPENED BAUDRATE:921600
Press Enter key to continue...
```

⑤ Key in “0” (0. exit/back)

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : UART
=====OTA Command Menu=====
Please choice following one item:
1.Get OTA Loader Version.....[OTA LOADER]
2.Read WE2 Chip id.....[OTA LOADER]
3.Getstatus (Get WE-II current status).....[OTA LOADER]
4.ReBoot WEII.....[OTA LOADER]
5.Busy pin select.....[OTA LOADER]
6.Read busy pin select.....[OTA LOADER]
7.Reserved.....[NONE]
8.Intoupg.....[APPLICATION]
0.exit/back

                               0
```

OTA Upgrade Operations (4/6)

⑥ Key in “2” (2. Upgrade)

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb 1 2024=====
Transfer Protocol : UART
=====OTA Main Menu=====
Please choice following one item:
1.Command
2.Upgrade
3.Others
0.exit
_____
                2
```

⑦ Key in “1” (1. UPGRADE ALL IMAGE(Upgrade all image file))

```
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb 1 2024=====
Transfer Protocol : UART
=====OTA Upgrade Menu=====
Please choice following one item:
1.UPGRADE ALL IMAGE(Upgrade all image file)
2.UPGRADE All APPLICATIONs
3.UPGRADE NS APPLICATIONs
0.exit/back
_____
                1_
```


OTA Upgrade Operations (5/6)

⑧ Key in “1” (1. output.img)

```
Upgrade Image
=====File List=====
Please select file:
1. output.img
0. exit
=====File List=====
1
```

⑨ OTA upgrade started ...

```
img_file = img\output.img
cstm_layout_file = img\cstm_layout.bin
Getstatus
status = 0x000c0000

==>OTA_STATUS_OTA_LOADER_IDLE
status = 0x000c0000

==>OTA_STATUS_OTA_LOADER_IDLE
hx_img_upgrade Upgrade Started
hx_memdsp->info.num_sbs = 0x6
part_idx = 0 pat_type = 0
start cmd : 0x0, 0x4ac0001, 0x15000, 0x14000, 0xbaa0
-count_length = 0x0
Getstatus
status = 0x00130000

==>OTA_STATUS_OTA_LOADER_INIT
status = 0x00130000

==>OTA_STATUS_OTA_LOADER_INIT
-count_length = 0x0
\count_length = 0x100
/count_length = 0x200
-count_length = 0x300
\count_length = 0x400
```

OTA Upgrade Operations (6/6)

⑩ OTA upgrade finished

```
==>OTA_STATUS_OTA_RECEIVE_UPGRADE_DATA_DONE  
.  
Burn flash success  
Getstatus  
status = 0x000c0000  
  
==>OTA_STATUS_OTA_LOADER_IDLE  
status = 0x000c0000  
  
==>OTA_STATUS_OTA_LOADER_IDLE  
ReBoot WE2  
Upgrade success Upgrade Finished  
Press Enter key to continue...
```